# Technology Adoption and Schooling as Complements: The Promise of School-Based Agricultural Education in Africa

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**Abstract**

In sub-Saharan Africa, wide heterogeneity across farms makes other households' experience a noisy guide to what suits a household's own farms, slowing technology diffusion, while low perceived returns to schooling drive students to drop out. We develop a model of why school-based agricultural education (SBAE), a holistic program in which trained science teachers integrate inquiry-based learning with student-led school demonstration farms and home farming projects, addresses this dual problem. Our model builds on the target-input framework, in which a household learns a technology's suitability only by using it in production or by observing others who do. We add a third channel: the more of a child's time the household devotes to schooling, the more precisely it learns from the school's production signals. Schooling and technology adoption become complements. First, schooling carries an option value on the farm: by raising the probability of future adoption, it makes attendance worthwhile even where labor-market returns are low. Second, SBAE effectively subsidizes information acquisition by delivering information through schooling, whose cost households already bear for its educational returns, countering the under-supply of information in the presence of learning spillovers. We provide experimental evidence from Liberia that SBAE cost-effectively raises both agricultural technology adoption and school attendance, and use extensions of the model to study how SBAE's design can be strengthened, including (i) how effectively inquiry-based learning raises the marginal informational return to schooling, and (ii) how effectively SBAE's home visits aggregate households' dispersed knowledge and feed it back to them.

# 1 Model

We use a simple generalization of the target-input model (Jovanovic and Nyarko, 1996; Foster and Rosenzweig, 1995) to frame the role of school-based agricultural education (SBAE) in the flow of information about a new technology, and in the incentives for schooling. The focus is on heterogeneity across farms and farmers: the optimal way of using a new technology varies across space and people, so information about one's own suitability is the scarce input. In the standard framework a household learns about that suitability in two ways: by using the technology in production, or by observing others who do. Heterogeneity limits the second channel, and the learning spillovers it creates lead each household to under-supply the first. SBAE adds a third channel, which runs through the child's schooling. The model lets us discuss the trade-offs and complementarities between school learning, future profits and current production, and between own experimentation and social learning. We refer to the farmer and her household interchangeably.

## 1.1 Technology and production

**Technology choice.** In each period $t$ farmer $i$ uses either the traditional technology ($T_{it} = 0$) or the new technology ($T_{it} = 1$). The traditional technology (planting flat, say) is well known to the farmer: it is risky, but she knows how to allocate her time and effort to get as much as she can from her farm. The new technology (planting in rows on ridges) is less familiar. She has beliefs about how best to use it on her own land, but knows her farm differs from others in many ways.

For now the adult supplies her full endowment of one unit of labor inelastically to her own farm, and that labor is embedded directly in the production function; child labor and schooling are introduced in Section 1.5. Output is

$$y_{it} = \begin{cases} f_0(\varepsilon_{0it}), & T_{it} = 0, \\ f_1(\omega_i + \varepsilon_{it}), & T_{it} = 1. \end{cases} \tag{1}$$

**Assumptions.**

- $f_0$ and $f_1$ are strictly increasing and strictly concave.
- $\omega_i$ is a permanent, farmer-specific measure of the appropriateness of the new technology for that farmer: a characteristic of the farmer and/or her land that determines how productive the new technology is on her plot. In reality $\omega_i$ is multidimensional; here we treat it as a scalar.
- *Heterogeneity across farms.* Suitability differs across farms. For any other plot $j$, whether a neighbor's farm or the school's, $\omega_j = \omega_i + d_{ij}$, where the mismatch $d_{ij} \sim N(0, \tau^2)$ is independent of $\omega_i$, of the other mismatches and of all production shocks. The parameter $\tau^2$ measures how much farms differ.[1]
- $\varepsilon_{0it}$ and $\varepsilon_{it}$ are i.i.d. transitory, farmer- and technology-specific shocks, each distributed $N(0, \sigma_\varepsilon^2)$ with $\sigma_\varepsilon^2$ known (idiosyncratic weather or pest variation). Write $\rho_\varepsilon \equiv 1/\sigma_\varepsilon^2$.
- Neither technology is completely dominant: there are values of $\omega_i$ for which $f_0(0) > f_1(\omega_i)$ and others for which the reverse holds.
- Farmers are risk neutral. This is of course counterfactual, but it makes the mechanism transparent; risk aversion and incomplete insurance are discussed separately.

[1] Treating mismatches as independent keeps updating additive across sources. When the mismatch between two given plots is permanent, repeated observation of the same plot cannot average it out, and the cumulative precision it can provide is bounded by $1/\tau^2$. We return to this structure in the extensions (Section 3).

**Beliefs.** The main focus of the model is that the farmer does not know $\omega_i$. She knows that it matters (she knows the general shape of $f_1$), and she holds a belief about it, which may come from a radio programme, conversations with other farmers, or a visit from an extension agent. At time $t$ her beliefs are

$$\omega_i \sim N\big(\mu_{it}, \sigma_{it}^2\big), \qquad \rho_{it} \equiv \frac{1}{\sigma_{it}^2}, \tag{2}$$

where $\mu_{it}$ is her best guess of the appropriateness of the technology for her own farm and $\rho_{it}$ is the strength, or precision, of that belief.

## 1.2 Expected output and the static adoption rule

Since the shocks have mean zero, expected output under the traditional technology is a constant,

$$\mathbb{E}[f_0(\varepsilon_{0it})] = \int f_0(\varepsilon)\, N(\varepsilon \mid 0, \sigma_\varepsilon^2)\, d\varepsilon \equiv g_0, \tag{3}$$

while expected output under the new technology is a function of the farmer's beliefs,

$$\mathbb{E}[f_1(\omega_i + \varepsilon_{it}) \mid \mu_{it}, \rho_{it}] = \int f_1(\omega+\varepsilon)\, N\big(\omega+\varepsilon \mid \mu_{it}, \sigma_{it}^2 + \sigma_\varepsilon^2\big)\, d(\omega+\varepsilon) \equiv g_1(\mu_{it}, \rho_{it}). \tag{4}$$

**Comparative statics.** Because $f_1$ is increasing and strictly concave in $\omega + \varepsilon$,

$$\frac{\partial g_1}{\partial \mu_{it}} > 0, \qquad \frac{\partial g_1}{\partial \rho_{it}} > 0. \tag{5}$$

The first is immediate. The second follows because $\omega$ and $\varepsilon$ are independent normals, so their sum is normal with variance equal to the sum of the variances; Jensen's inequality then ensures that reducing the variance of that sum raises the expected value of $f_1$. A sufficiently optimistic, or a sufficiently confident, farmer expects at least as much from the new technology as from the traditional one.

**Thresholds.** For any fixed $\rho_{it}$ there is a critical mean $\mu_{it}^*$ with $g_1(\mu_{it}^*, \rho_{it}) = g_0$, such that $g_1 > g_0$ for all $\mu_{it} > \mu_{it}^*$ and $g_1 < g_0$ for all $\mu_{it} < \mu_{it}^*$. Symmetrically, for any fixed $\mu_{it}$ there is a precision threshold $\rho_{it}^*$ with the analogous property. If the model were static, a risk neutral farmer would use the new technology if and only if $g_1(\mu_{it}, \rho_{it}) \geq g_0$. Once she can learn, the adoption decision is different.

## 1.3 Learning by doing and learning from others: Bayesian updating

**Learning by doing.** Suppose the farmer uses the new technology in period $t$ and observes output $y_{it}$. Because $f_1$ is strictly monotone, she can invert the production function and recover a noisy signal of her own $\omega_i$:

$$s_{it}^f = \omega_i + \varepsilon_{it} = f_1^{-1}(y_{it}), \qquad \varepsilon_{it} \sim N(0, \sigma_\varepsilon^2). \tag{6}$$

The transitory shock obscures the signal: the greater its variance, the less information the harvest conveys. The precision of the signal is $\rho_\varepsilon = 1/\sigma_\varepsilon^2$, which is known.

**Learning from others.** Suppose in addition that $n_{it}$ of the farmer's neighbors use the new technology in period $t$, and that she observes their harvests through conversation or direct observation. Neighbor $j$'s harvest reveals $s^N_{jt} = \omega_j + \varepsilon_{jt} = \omega_i + d_{ij} + \varepsilon_{jt}$. As a signal of the farmer's *own* suitability, it has variance $\tau^2 + \sigma^2_\varepsilon$ and hence precision

$$\rho_N \equiv \frac{1}{\tau^2 + \sigma^2_\varepsilon} < \rho_\varepsilon. \tag{7}$$

A neighbor's experience is a noisier guide than one's own, and it is noisier the more farms differ: $\rho_N$ is decreasing in $\tau^2$. We presume this social learning is free of effort, though the effort involved in improving its precision could be modelled.

**Updating.** Given the prior (2), the posterior is $N(\mu_{i,t+1}, \sigma^2_{i,t+1})$ with

$$\mu_{i,t+1} = \frac{\rho_{it}\mu_{it} + T_{it}\rho_\varepsilon s^f_{it} + \rho_N \sum_{j=1}^{n_{it}} s^N_{jt}}{\rho_{it} + T_{it}\rho_\varepsilon + n_{it}\rho_N}, \tag{8}$$

$$\rho_{i,t+1} = \rho_{it} + T_{it}\rho_\varepsilon + n_{it}\rho_N. \tag{9}$$

The new mean is a precision-weighted average of the prior mean and the new signals, and the posterior precision is strictly higher than the prior precision whenever the farmer observes any harvest. Note that the farmer learns only if the technology is used in production, on her own farm or on her neighbors': if $T_{it} = 0$ and $n_{it} = 0$ she observes nothing about $\omega_i$ and $(\mu_{i,t+1}, \rho_{i,t+1}) = (\mu_{it}, \rho_{it})$.

## 1.4 The dynamic problem

We first study a farmer who learns only from her own use ($n_{it} = 0$), and then return to learning from others. Let $\beta < 1$ be the discount factor. The discounted value of expected future output is

$$V(\mu_{it}, \rho_{it}) = \mathbb{E}\left( \sum_{\tau=t}^{\infty} \beta^{\tau-t} \max_{T_{i\tau}\in\{0,1\}} \Big[ (1 - T_{i\tau}) g_0 + T_{i\tau} g_1(\mu_{i\tau}, \rho_{i\tau}) \Big] \;\Big|\; \mu_{it}, \rho_{it} \right). \tag{10}$$

Breaking the sum into today and the future, and using the law of iterated expectations, gives the recursive form

$$V(\mu_{it}, \rho_{it}) = \max_{T_{it}\in\{0,1\}} (1 - T_{it}) g_0 + T_{it} g_1(\mu_{it}, \rho_{it}) + \beta\, \mathbb{E}\big[ V(\mu_{i,t+1}, \rho_{i,t+1}) \mid \mu_{it}, \rho_{it} \big]. \tag{11}$$

In the current period she weighs expected output this season (the first two terms) against the effect of today's choice on the future flow of output. In this model the only channel through which today's choice affects the future is beliefs, via (8)–(9).

**Non-adoption is absorbing.** If $T_{it} = 0$ she receives $g_0$ and no signal, so $\mu_{i,t+1} = \mu_{it}$ and $\rho_{i,t+1} = \rho_{it}$: she is in period $t+1$ exactly where she was in period $t$. Hence if it is optimal not to use the new technology at $t$, it remains optimal at $t+1$, and

$$V(\mu_{it}, \rho_{it}) = g_0 + \sum_{s=1}^{\infty} \beta^s g_0 \equiv V^0, \tag{12}$$

the present discounted value of the traditional technology forever. Learning by doing alone thus has the strong implication that $T_{it} = 0$ is an absorbing state.

**The adoption condition.** The farmer uses the new technology if and only if

$$g_1(\mu_{it},\rho_{it}) + \beta\,\mathbb{E}[V(\mu_{i,t+1},\rho_{i,t+1}) \mid \mu_{it},\rho_{it}] \;\geq\; g_0 + \beta V^0,$$

or, rearranged,

$$\underbrace{g_0 - g_1(\mu_{it},\rho_{it})}_{\text{current cost of experimenting}} \;\leq\; \underbrace{\beta\,\mathbb{E}[V(\mu_{i,t+1},\rho_{i,t+1}) \mid \mu_{it},\rho_{it}] - V^0}_{\text{option value of learning}}. \tag{13}$$

Since $V(\mu,\rho) \geq V^0$ always (the farmer can always revert), $g_1 \geq g_0$ is sufficient for adoption but not necessary. The concavity of $f_1$ implies $g_1(\mu,\rho+\rho_\varepsilon) > g_1(\mu,\rho)$ and hence $V(\mu,\rho+\rho_\varepsilon) > V(\mu,\rho)$: the farmer values the learning itself, and will accept an expected loss in current output to obtain it. Both comparative statics carry over to the value function,

$$\frac{\partial V}{\partial \mu_{it}} \geq 0, \qquad \frac{\partial V}{\partial \rho_{it}} \geq 0, \tag{14}$$

with strict inequality when $T_{it} = 1$.

**Abandonment.** The left-hand side of (13) is decreasing in $\mu_{it}$ and the right-hand side is increasing in $\mu_{it}$, so for a given precision there is a cutoff $\bar{\mu}_i$, independent of $t$, such that (13) holds for all $\mu_{it} \geq \bar{\mu}_i$ and fails for $\mu_{it} < \bar{\mu}_i$. The farmer becomes indifferent between continuing and stopping when she observes a sufficiently bad signal about suitability. Because the updating rule (8) is linear, the cutoff in beliefs maps one-for-one into a cutoff in the signal, and hence in observed output:

$$\bar{s}_i = \frac{(\rho_{it}+\rho_\varepsilon)\bar{\mu}_i - \rho_{it}\mu_{it}}{\rho_\varepsilon}, \qquad \bar{y}_i = f_1(\bar{s}_i), \tag{15}$$

so that $\mu_{i,t+1} \geq \bar{\mu}_i$ if and only if $y_{it} \geq \bar{y}_i$. From the farmer's period-$t$ perspective the signal is distributed $s^f_{it} \sim N(\mu_{it}, \sigma^2_{it} + \sigma^2_\varepsilon)$, so the probability that she abandons the new technology after this season is

$$p^{\text{abandon}}_{it} = \Pr\!\left(s^f_{it} < \bar{s}_i\right) = \Phi\left(\frac{\bar{s}_i - \mu_{it}}{\sqrt{\sigma^2_{it} + \sigma^2_\varepsilon}}\right). \tag{16}$$

Writing the continuation value in these terms makes (13) explicit: an experimenting farmer continues with probability $1 - p^{\text{abandon}}_{it}$ and reverts to $V^0$ otherwise, so

$$g_0 - g_1(\mu_{it},\rho_{it}) \leq \beta\left[\int_{\bar{s}_i}^{\infty} V\!\left(\tfrac{\rho_{it}\mu_{it}+\rho_\varepsilon s}{\rho_{it}+\rho_\varepsilon},\, \rho_{it}+\rho_\varepsilon\right)\phi\left(s;\mu_{it},\sigma^2_{it}+\sigma^2_\varepsilon\right)ds + p^{\text{abandon}}_{it}V^0 - V^0\right]. \tag{17}$$

**Learning from others and the under-supply of information.** When neighbors use the technology ($n_{it} > 0$), non-adoption is no longer absorbing: a farmer who does not experiment still updates by (8)–(9). But this channel depends entirely on others' willingness to experiment, and each experimenter's harvest is also information for her neighbors. Suppose household $i$'s harvest is observed by a set $\mathcal{N}_i$ of neighbors. If $i$ uses the technology, each $k \in \mathcal{N}_i$ gains a signal of precision $\rho_N$, worth

$$\Delta_k \equiv \beta\left(\mathbb{E}\big[V(\mu'_k, \rho'_k + \rho_N)\big] - \mathbb{E}\big[V(\mu'_k,\rho'_k)\big]\right) \;\geq\; 0, \tag{18}$$

where the inequality holds because more information never lowers the value of a decision problem (Blackwell, 1953), and is strict whenever the signal changes $k$'s adoption decision with positive probability.

**Lemma 1** (Under-supply of information)**.** *Household i experiments if and only if* (13) *holds. A planner who values all households' output experiments at i if and only if*

$$g_0 - g_1(\mu_{it}, \rho_{it}) \;\le\; \beta\, \mathbb{E}[V(\mu_{i,t+1}, \rho_{i,t+1})] - V^0 + \sum_{k \in \mathcal{N}_i} \Delta_k.$$

*Whenever some neighbor has $\Delta_k > 0$, the set of beliefs at which i experiments is strictly smaller in the decentralized outcome than in the planner's: information is under-supplied.*

*Proof.* The household's condition is (13). The planner's condition adds the value to each neighbor of the signal that $i$'s harvest generates, $\sum_k \Delta_k \geq 0$, which $i$ does not internalize. Since the left-hand side is decreasing in $\mu_{it}$, the planner's cutoff in $\mu_{it}$ lies strictly below $\bar{\mu}_i$ whenever $\sum_k \Delta_k > 0$. □

Three lessons follow:

1. *Learning externality.* Because experimenters do not internalise the value of the signals they generate for others, too few farmers choose $T = 1$ (Lemma 1).

2. *Strategic delay.* The value of learning from one's own experience is reduced when others may experiment first, reinforcing the first effect (Foster and Rosenzweig, 1995; Bandiera and Rasul, 2006).

3. *Heterogeneity.* The value of learning from others falls with $\tau^2$, by (7). The more farms differ, the less a neighbor's experience reveals, and the more each household must rely on costly experimentation on its own land (Conley and Udry, 2010).

### 1.5 School or farm work?

We now enrich the model with the trade-offs that shape schooling. A child devotes a share $e_{sit}$ of her time to school; the more time she devotes, the more she learns and the higher her eventual wage, whose present discounted value we write $S(e_{sit})$, with $S' > 0$ and $S$ concave. Where labor-market returns to schooling are low, or are perceived to be low (Jensen, 2010), $S'$ is small. Alternatively she supplies $1 - e_{sit}$ units of labor to the farm. To accommodate this trade-off we make the role of labor in farm production explicit; it was not needed above, because the adult supplies labor inelastically. Both technologies can use child labor:

$$y_{it} = \begin{cases} f_0\big(1 - e_{sit}; \varepsilon_{0it}\big), & T_{it} = 0, \\ f_1\big(1 - e_{sit}; \omega_i + \varepsilon_{it}\big), & T_{it} = 1, \end{cases} \tag{19}$$

with expected outputs

$$\mathbb{E}[y_{it} \mid e_{sit}] = \int f_0(1 - e_{sit}; \varepsilon)\, N(\varepsilon \mid 0, \sigma_\varepsilon^2)\, d\varepsilon \equiv g_0(e_{sit}), \tag{20}$$

$$\mathbb{E}[y_{it} \mid e_{sit}, \mu_{it}, \rho_{it}] = \int f_1(1 - e_{sit}; \omega + \varepsilon)\, N\big(\omega + \varepsilon \mid \mu_{it}, \sigma_{it}^2 + \sigma_\varepsilon^2\big)\, d(\omega + \varepsilon) \equiv g_1(e_{sit}; \mu_{it}, \rho_{it}). \tag{21}$$

These make explicit one of the costs of schooling: lost output under either technology. $g_1(e_{sit}; \mu_{it}, \rho_{it})$ is concave in $1 - e_{sit}$ and, as before, increasing in $\mu_{it}$ and $\rho_{it}$. Write

$$w_{it} \equiv (1 - T_{it})\, g_0'(1 - e_{sit}) + T_{it}\, g_1'(1 - e_{sit}; \mu_{it}, \rho_{it}) \tag{22}$$

for the marginal product of child labor on the farm. By concavity, $w_{it}$ is non-decreasing in $e_{sit}$.

**Schooling without SBAE.** Absent any informational role for school, schooling simply balances the marginal product of child labor on the farm against the marginal return to education:

$$w_{it} = S'(e_{sit}). \tag{23}$$

Denote the solution $e^0_{it}$. Schooling therefore depends on the technology in use only through the marginal product of child labor, and there is no reason for an adopting and a non-adopting household to school their children differently beyond that. Where $S'(0) \leq w_{it}$ at $e_{sit} = 0$, the child does not attend at all.

### 1.6 SBAE: learning from the school's production signals

SBAE is a holistic program in which trained science teachers integrate inquiry-based learning with student-led school demonstration farms and home farming projects. Students run the new technology in production on the school's demonstration farm, and apply it on a small home farming project on their family's land. SBAE can affect adoption through two channels.

1. *Belief shifts.* The parent may be exposed to messages, transmitted via the student or directly from the teacher, that change her priors: participation raises $\mu_{it}$ or raises $\rho_{it}$, making adoption weakly more attractive.

2. *The school's production signals.* When the child participates in school-based farming, she observes noisy signals of the appropriateness of the technology, which she brings home. These signals come from production that the school, not the household, undertakes. Crucially, this means farmers learn about the new technology even if neither they nor their neighbors are using it.

We focus on the second channel. The school generates two production signals.

**The demonstration farm.** The demonstration farm is one more plot in the catchment, with suitability $\omega_c = \omega_i + d_{ic}$. The more time the student devotes to school, the more precisely she learns what the demonstration farm reveals: she observes $\tilde{\omega}_{st} = \omega_c + \varepsilon_{st}$ with $\varepsilon_{st} \sim N(0, 1/\pi_d(e_{sit}))$ and $\pi'_d > 0$. As a signal of the household's own suitability, it has precision

$$\rho_d(e_{sit}) = \frac{\pi_d(e_{sit})}{1 + \tau^2 \pi_d(e_{sit})} < \frac{1}{\tau^2}. \tag{24}$$

Like a neighbor's harvest, the demonstration farm is a guide to the household's own land only to the extent that the two plots are alike.

**The home farming project.** The home farming project applies the technology, at negligible scale, on the household's own land. It yields $\omega_{ht} = \omega_i + \eta_{ht}$ with precision $\rho_h(e_{sit})$, $\rho'_h > 0$. Because it is generated on the household's own farm, it carries no mismatch term, and because its scale is negligible, it puts no meaningful share of the harvest at risk.

**Schooling time.** In the baseline model we treat all time the child devotes to SBAE (in the classroom, on the demonstration farm and on the home farming project) as schooling, valued through $S(e_{sit})$. How effectively SBAE's activities raise the informational return to that time is one of the design questions we take up in Section 3. The total precision of the school's production signals is

$$\rho_s(e_{sit}) \equiv \rho_d(e_{sit}) + \rho_h(e_{sit}), \qquad \rho_s(0) = 0, \quad \rho'_s > 0, \quad \rho''_s \leq 0, \tag{25}$$

and we write $\omega^s_i$ for the precision-weighted average of the two signals, which has precision $\rho_s(e_{sit})$ about $\omega_i$.

**Updating with SBAE.** Combining own experience (available only when $T_{it} = 1$), others' experience (available when $n_{it} > 0$) and the school's production signals (available whenever the child attends), beliefs evolve as

$$\mu_{i,t+1} = \frac{\rho_{it}\mu_{it} + T_{it}\rho_\varepsilon s^f_{it} + \rho_N \sum_{j=1}^{n_{it}} s^N_{jt} + \rho_s(e_{sit})\,\omega^s_t}{\rho_{it} + T_{it}\rho_\varepsilon + n_{it}\rho_N + \rho_s(e_{sit})}, \tag{26}$$

$$\rho_{i,t+1} = \rho_{it} + T_{it}\rho_\varepsilon + n_{it}\rho_N + \rho_s(e_{sit}). \tag{27}$$

**The value function with schooling and SBAE.**

$$V(\mu_{it},\rho_{it}) = \max_{e_{sit},\,T_{it}\in\{0,1\}} (1-T_{it})g_0(e_{sit}) + T_{it}g_1(e_{sit};\mu_{it},\rho_{it}) + S(e_{sit}) + \beta\,\mathbb{E}\big[V(\mu_{i,t+1},\rho_{i,t+1}) \mid e_{sit},\mu_{it},\rho_{it}\big], \tag{28}$$

with $V$ increasing in both $\mu$ and $\rho$. Let

$$\lambda_{it} \equiv \beta\,\mathbb{E}\left[\frac{\partial V(\mu_{i,t+1},\rho_{i,t+1})}{\partial \rho}\right] \geq 0 \tag{29}$$

denote the marginal value of precision. The optimal schooling time now reflects the learning value of SBAE as well as the value of child labor on the farm and the human capital return to education:

$$w_{it} = S'(e_{sit}) + \beta\,\mathbb{E}\left[\frac{\partial V}{\partial \rho}\frac{\partial \rho_s}{\partial e_s} + \frac{\partial V}{\partial \mu}\frac{\partial \mu_{i,t+1}}{\partial e_s}\right] = S'(e_{sit}) + \lambda_{it}\,\rho_s'(e_{sit}). \tag{30}$$

The second equality follows because the expected effect of schooling time through the posterior mean vanishes:

$$\mathbb{E}\left[\frac{\partial V(\mu_{i,t+1},\rho_{i,t+1})}{\partial \mu}\frac{\partial \mu_{i,t+1}}{\partial e_s}\right] = \int \frac{\partial V}{\partial \mu}\frac{\partial \mu_{i,t+1}}{\partial e_s} N(\omega \mid \mu_{it},\rho_{it})\,d\omega = 0, \tag{31}$$

by the local linearity of $V$: a more precise signal is not, in expectation, a more favourable one. What remains is a non-negative information-gain term.

## 1.7 Schooling and technology adoption as complements

**Definition 1.** Schooling and technology adoption are *complements* if (a) more schooling raises the probability that the household adopts the new technology, and (b) a more valuable adoption option raises the household's demand for schooling.

Without SBAE neither holds: by (23), schooling responds to the technology only through the marginal product of child labor, and schooling has no effect on what the household learns. With SBAE both hold.

**Adoption by non-adopters.** The sharpest departure from Section 1.4 is that $T_{it} = 0$ is no longer absorbing, even when no neighbor experiments. Without using the technology the farmer observes $\omega^s_t$ and updates. Take a non-adopter with $n_{it} = 0$. For given $(\mu_{it},\rho_{it})$, adoption occurs at $t+1$ if $\mu_{i,t+1} \geq \bar{\mu}_i$, which by (26) with $T_{it} = 0$ requires a school signal above

$$\omega^*_t = \frac{(\rho_{it} + \rho_s(e_{sit}))\,\bar{\mu}_i - \rho_{it}\mu_{it}}{\rho_s(e_{sit})}, \tag{32}$$

so that the probability of adoption in the next period is

$$p_{it}^{T=1} = \int_{\omega_t^*}^{\infty} \phi\left(\omega; \mu_{it}, \sigma_{it}^2 + \sigma_s^2(e_{sit})\right) d\omega = 1 - \Phi\left( (\bar{\mu}_i - \mu_{it}) \sqrt{\frac{\rho_{it}\left(\rho_{it} + \rho_s(e_{sit})\right)}{\rho_s(e_{sit})}} \right), \tag{33}$$

where $\sigma_s^2(e_{sit}) = 1/\rho_s(e_{sit})$ and $d\sigma_s^2/de_s < 0$.

**Proposition 1** (Schooling and adoption are complements)**.**

*(a)* Schooling raises adoption. *For a non-adopting household with $\mu_{it} < \bar{\mu}_i$, the probability of adoption in the next period, $p_{it}^{T=1}$, is strictly increasing in schooling time $e_{sit}$, holding the adoption cutoff $\bar{\mu}_i$ fixed. As $\rho_s \to \infty$ it tends to $1 - \Phi\big((\bar{\mu}_i - \mu_{it})\sqrt{\rho_{it}}\big)$, the prior probability that the technology in fact suits the household's land.*

*(b)* Schooling carries an option value on the farm. *The marginal value of precision $\lambda_{it}$ is strictly positive if and only if the household adopts with positive probability at some future date. Whenever $\lambda_{it} > 0$, schooling under SBAE strictly exceeds schooling without it, $e_{it}^* > e_{it}^0$ (when $e_{it}^0 < 1$), for adopters and non-adopters alike.*

*(c)* Attendance where labor-market returns are low. *A household whose child would not attend without SBAE, because $S'(0) \le w_{it}(0)$, sends the child to school under SBAE whenever*

$$S'(0) + \lambda_{it}\, \rho_s'(0) > w_{it}(0).$$

*(d)* The value of the adoption option raises schooling. *At an interior optimum,*

$$\frac{de_{it}^*}{d\lambda_{it}} = \frac{\rho_s'(e_{it}^*)}{\partial w_{it}/\partial e_s - S''(e_{it}^*) - \lambda_{it}\rho_s''(e_{it}^*)} > 0.$$

*Proof.* (a) By (33), $p_{it}^{T=1} = 1 - \Phi(z)$ with $z = (\bar{\mu}_i - \mu_{it})\sqrt{\rho_{it}(1 + \rho_{it}/\rho_s)}$. With $\bar{\mu}_i > \mu_{it}$, $z > 0$ is strictly decreasing in $\rho_s$, which is strictly increasing in $e_{sit}$. The limit follows directly. (b) If the household adopts with probability zero in every future state, its continuation value is $V^0$ for all beliefs and $\partial V/\partial \rho = 0$. If it adopts with positive probability, additional precision raises $g_1$ in those states by (5), so $\lambda_{it} > 0$. Comparing (30) with (23), the marginal benefit of schooling is higher by $\lambda_{it}\rho_s' > 0$ at every $e_{sit}$, while the marginal cost $w_{it}$ is unchanged; since the marginal benefit is decreasing and the marginal cost non-decreasing in $e_{sit}$, the solution rises. (c) is the corner condition for (30) at $e_{sit} = 0$. (d) follows from implicit differentiation of (30); the denominator is positive because $w_{it}$ is non-decreasing in $e_{sit}$, $S'' \le 0$ and $\rho_s'' \le 0$. □

Part (a) is the first direction of the complementarity: schooling raises adoption, and it does so for exactly the households that learning by doing leaves stuck, those that are too pessimistic to experiment. If the cutoff $\bar{\mu}_i$ is non-increasing in precision, as it is in the static rule (5), the effect is reinforced, because a more precise household also needs a less favourable signal to adopt. Parts (b) to (d) are the second direction. Schooling acquires an option value on the farm: it raises the probability of future adoption, so its value rises with the value of the adoption option itself. Attendance becomes worthwhile even where labor-market returns are low, and it is the prospect of adoption, not adoption today, that makes it so.

## 1.8 SBAE as a subsidy to information acquisition

Without SBAE, a household that wants information about its own suitability has two sources. It can experiment, at a current cost of $g_0 - g_1(\mu_{it}, \rho_{it})$ in expected output for precision $\rho_\varepsilon$, or it can wait for its neighbors to experiment, which they do too rarely (Lemma 1). SBAE adds a third source, schooling, whose cost the household already bears for a different reason.

To see this, compare the cost of a unit of precision obtained through the school with its resource cost. An additional unit of schooling time costs $w_{it}$ in lost farm output and yields $\rho_s'(e_{sit})$ units of precision, so the resource cost of a unit of precision is $w_{it}/\rho_s'(e_{sit})$. But the household values the same time for its educational return $S'(e_{sit})$. The private cost of a unit of precision is therefore

$$c_{it}^s = \frac{w_{it} - S'(e_{sit})}{\rho_s'(e_{sit})} \;<\; \frac{w_{it}}{\rho_s'(e_{sit})}. \tag{34}$$

For comparison, the cost of a unit of precision through own experimentation is $c_{it}^T = \big(g_0 - g_1(\mu_{it}, \rho_{it})\big)/\rho_\varepsilon$, with no offsetting return.

**Proposition 2** (SBAE subsidizes information acquisition)**.**

*(a)* Information is bundled with schooling the household already buys. *If $e_{it}^0 > 0$, the household obtains precision $\rho_s(e_{it}^0) > 0$ about its own suitability at no cost beyond the schooling it would choose without SBAE.*

*(b)* The subsidy at the margin. *Beyond $e_{it}^0$, the private cost of precision is below its resource cost by $S'(e_{sit})/\rho_s'(e_{sit})$ per unit, an effective subsidy at rate $S'(e_{sit})/w_{it}$. The subsidy is financed by the educational returns to schooling, not by transfers.*

*(c)* Countering the under-supply of information. *The school's production signals reach the household whether or not it, or any neighbor, uses the technology. For every household, posterior precision rises by at least $\rho_s(e_{it}^0)$, including households that do not experiment and whose neighbors do not either, which are the households for which the under-supply of Lemma 1 binds.*

*Proof.* (a) At $e_{it}^0$, $S'(e_{it}^0) = w_{it}$, so the household would choose $e_{it}^0$ without any informational motive, and by (27) that schooling delivers $\rho_s(e_{it}^0)$. (b) follows from (34). (c) follows from (27), since $\rho_s(e_{sit})$ enters independently of $T_{it}$ and $n_{it}$, and from $e_{it}^* \geq e_{it}^0$ (Proposition 1). □

The proposition is the second source of the complementarity. Learning spillovers leave households with too little information because the only way to produce it, using the technology in production, is costly to the household that bears it and valuable to others. SBAE does not remove the externality; it moves the production of information into an institution whose time the household is already willing to pay for. Two features of the school's signals reinforce this. The demonstration farm is a single experiment observed by every enrolled household, so its cost does not grow with the number of households it informs. And since the school does not wait for anyone to experiment first, its signals are not subject to strategic delay.

## 1.9 Implications

The marginal value of schooling under SBAE includes an extra, non-negative information term, and this is the sense in which the program ties agricultural education to schooling rather than simply adding one to the other.

- If $T_{it} = 1$, more schooling raises the precision of the signals from the demonstration farm and from the home farming project, which strictly increases the continuation value.

- If $T_{it} = 0$, more schooling still raises the probability of crossing the adoption threshold in the future, per (33), and so raises the option value of education. Non-adoption ceases to be absorbing, even where no neighbor experiments.
- Because the school's production signals arrive whether or not the farmer or her neighbors experiment, and because their cost is largely borne for schooling's own returns, SBAE supplies information that decentralized experimentation under-provides.

Thus SBAE generates an additional motive for schooling: by tying attendance to information acquisition, it can raise schooling even when the new technology is not adopted immediately, and it can raise adoption even among households whose children's time is valuable on the farm.

**What the baseline model leaves out.** Several features of SBAE are outside the baseline model. Leadership training may raise the precision with which students convey what they learn to their parents, which would scale up $\rho_s$. Parents may observe the demonstration farm directly, which adds a component of precision that does not depend on $e_{sit}$. When mismatches between plots are permanent, the demonstration farm and neighbors' harvests can teach a household at most $1/\tau^2$ about its own land over any horizon, while the home farming project faces no such bound. Finally, the household is risk neutral and the adoption decision is binary.

## 2 Experimental evidence from Liberia

*[To be written: the RCT design, effects of SBAE on agricultural technology adoption and on school attendance, and cost-effectiveness.]*

## 3 Extensions: strengthening SBAE's design

*[To be written.]* We plan two extensions.

(i) *Inquiry-based learning.* How effectively inquiry-based learning raises the marginal informational return to schooling, $\rho_s'(e_{sit})$, for example by teaching students to compare the technology against a control so that their home farming projects separate the technology's effect from the season's.

(ii) *Aggregating dispersed knowledge.* How effectively SBAE's home visits collect what households have learned on their own land, pool it across the catchment, and feed it back to them, including whether the pooled knowledge returns through the household or through the classroom.

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